Yongkang Li

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Research Interests: NLP, Information Retrieval, Robust Language Model Embeddings, Recommender systems

Education

- **University of Amsterdam, Informatics Institute**

  Lab: *Information Retrieval Lab* **PhD**, Oct 2023–Jun 2027(expected), Amsterdam, The Netherlands

  Supervisors: Prof. dr. Evangelos Kanoulas and dr. Panagiotis Eustratiadis

  ELLIS PhD Co-Supervisor: Prof. dr. Emine Yilmaz, University College London (UCL)

- **Southern University of Science and Technology, Department of Computer Science and Engineering**

  Lab: *SUSTech-UTokyo Joint Research Center on Super Smart City* **Master**, 2020–2023, Shenzhen, China

  Supervisors: Prof. dr. Xuan Song and Prof. dr. Zipei Fan

- **Beijing University of Posts and Telecommunications, School of Information and Communication Engineering**

  Lab: *Pattern Recognition and Intelligent Systems Laboratory* **Bachelor**, 2016–2020, Beijing, China

Main Research Projects

- **Embedding Robustness and Compression for Dense Retrieval**
  Embedding-based retrieval is widely deployed, yet dense retrievers remain fragile under distribution shifts and adversarial manipulation. My PhD research focuses on the robustness and efficiency of embedding-based retrieval, with two complementary research directions:
  I investigate the generalizability and stability of dense retrievers under distribution shifts, query perturbations, and corpus poisoning. In my recent **TOIS** submission, I systematically evaluate LLM-based dense retrievers and identify robustness gaps that persist despite improvements in model scale and retrieval effectiveness. My earlier work developed efficient corpus poisoning attacks and practical adversarial training methods, published at **SIGIR 2025** and **ECIR 2025**, with the latter receiving the **Best Reproducibility Paper Award**.
  In parallel, I develop post-hoc embedding compression methods that reduce dimensionality while preserving retrieval effectiveness, without retraining the encoder. I proposed Spectral Tempering (**SIGIR 2026**), which derives compression weights from corpus spectral statistics. In my subsequent work (under review at **WWW 2027**), I introduce relevance-guided residual calibration, using limited relevance judgments to refine these weights through direction-specific corrections.
- **Heterogeneous Graph Learning for Location-based Social Network Friend Recommendation**
  This research, part of my master's studies, focuses on friend recommendation in location-based social networks (LBSN) using user trajectory data. We explored the impact of social network relationships on human mobility, particularly long-distance travel. Our approach used heterogeneous multi-graphs and hypergraphs, with novel neural network models for learning and recommendation. The solution achieved state-of-the-art results on multiple real-world datasets and has been published in **TKDD**, **CIKM**, and **WWWJ**.

Industry Experience

Amazon.com, Inc. - Alexa for Shopping (Rufus Science) London, United Kingdom
Applied Scientist Intern (Manager: Gabriella Kazai) Sep 2026–Present

- **Low-Latency Personalized Shopping Page Generation**
  Developing personalized shopping pages for Alexa for Shopping, with tailored product information and comparisons. Researching retrieval-based methods to reduce LLM generation latency.

Temu Inc. - Research and Development Department Shanghai, China
Machine Learning Algorithm Engineer Jul 2023–Sep 2023

- **Conversational Intent Recognition**
  Developed an automated customer intent classification and analytics system to identify real-time user requests (e.g., returns, refunds, and shipping inquiries) during e-commerce customer service interactions. Engineered a hybrid model integrating deep learning architectures with traditional feature-matching techniques, fusing cleaned multi-turn dialogue data, customer purchase history, and real-time order status. Deployed the pipeline to accelerate agent response times and provide actionable statistical insights for cross-departmental business analysis.

Technical Skills

English:
- TOEFL-107, Reading 29, Listening 29, Speaking 23, Writing 26

Programming:
- Python, C++, Linux, Git; strong software engineering and coding practices.
- ML/DS stack: PyTorch, Pandas, NumPy, Scikit-learn, TensorFlow.
- HPC: proficient with Slurm for job scheduling, GPU/CPU resource allocation, and large-scale experimentation.

Selected Publications

Papers - First Author or Co-first Author (* indicates equal contribution)

- **Yongkang Li**, Panagiotis Eustratiadis, and Evangelos Kanoulas. Relevance-Guided Residual Calibration for Spectral Compression of Dense Retrieval Embeddings. Under review at the Web Conference 2027 (**WWW'2027**). [C]
- Dan Li, **Yongkang Li***, Simon Huesgen, and Hosein Azarbonyad, *Context-Enriched Figure Indexing for Scientific Multi-Modal RAG: An Industry Case Study at Scale*. [C]// In the 2026 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (**SIGIR-AP 2026**)
- **Yongkang Li**, Panagiotis Eustratiadis, Yixing Fan, and Evangelos Kanoulas. *On the Robustness of LLM-Based Dense Retrievers: A Systematic Analysis of Generalizability and Stability*. Under review at ACM Transactions on Information Systems (**TOIS**). Major Revision. Submitted: 08 April 2026. [C] [Paper]
- **Yongkang Li**, Panagiotis Eustratiadis, and Evangelos Kanoulas. Spectral Tempering for Efficient Embedding Compression in Dense Passage Retrieval. In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C] [Paper]
- Arne Eichholtz, **Yongkang Li***, Jutte Vijverberg, Tobias Groot and Mohammad Aliannejadi. *Hypencoder Revisited: Reproducibility and Analysis of Non-Linear Scoring for First-Stage Retrieval*.In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C] [Paper]
- **Yongkang Li**, Panagiotis Eustratiadis, Simon Lupart, and Evangelos Kanoulas. *Unsupervised Corpus Poisoning Attacks in Continuous Space for Dense Retrieval*. In the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2025**). [C] [Paper]
- **Yongkang Li**, Panagiotis Eustratiadis, and Evangelos Kanoulas. *Reproducing HotFlip for Corpus Poisoning Attacks in Dense Retrieval*. In the 47th European Conference on Information Retrieval (**ECIR'2025**). **Best Reproducibility Paper Award**. [C] [Paper]
- Dominykas Seputis, **Yongkang Li***, Karsten Langerak, and Serghei Mihailov. *Rethinking the Privacy of Text Embeddings: A Reproducibility Study of "Text Embeddings Reveal (Almost) As Much As Text"*. In the 19th ACM Conference on Recommender Systems **(RecSys'2025)**. [C] [Paper]
- **Yongkang Li**, Zipei Fan, and Xuan Song. *Heterogeneous Hyperbolic Hypergraph Neural Network for Friend Recommendation in Location-based Social Network*. ACM Transactions on Knowledge Discovery from Data (**TKDD**), 2024. [J] [Paper]
- **Yongkang Li**, Zipei Fan, Jixiao Zhang, Dengheng Shi, Tianqi Xu, Du Yin, Jinliang Deng, and Xuan Song. *Heterogeneous Hypergraph Neural Network for Friend Recommendation with Human Mobility*. In the 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**), 2022. [C] [Paper]
- **Yongkang Li**, Zipei Fan, Du Yin, Renhe Jiang, Jinliang Deng, and Xuan Song. *HMGCL: Heterogeneous Multigraph Contrastive Learning for LBSN Friend Recommendation*. World Wide Web (**WWWJ**), 2022. [J] [Paper]

Papers - Co-Author

- Kidist Amde Mekonnen, **Yongkang Li**, Yubao Tang, Simon Lupart and Maarten de Rijke. *Lost in Decoding? Reproducing and Stress-Testing the Look-Ahead Prior in Generative Retrieval*. In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C] [Paper]
- Yuxi Lin, **Yongkang Li**, Jie Xing, and Zipei Fan. *Multifaceted Scenario-Aware Hypergraph Learning for Next POI Recommendation*. In the 40th AAAI Conference on Artificial Intelligence (**AAAI'2026**). [C] [Paper]
- Peter Atandoh, **Yongkang Li**, Weikang Guo, Jingyu Guo, Jie Zou. *Less is More: Adaptive Prompt Compression and Exemplar Selection for Efficient Few-Shot Sentiment Analysis*. Expert Systems with Applications (**ESWA**). [J] [Paper]
- Jixiao Zhang, **Yongkang Li**, Ruotong Zou, Jingyuan Zhang, Renhe Jiang, Zipei Fan, and Xuan Song. *Hyper-relational Knowledge Graph Neural Network for Next POI Recommendation*. World Wide Web (**WWWJ**), 2024. [J] [Paper]

Self-evaluation

I am intellectually curious and proactive in exploring new ideas. I work effectively under pressure, take ownership of deliverables end-to-end, and collaborate well in teams with a strong sense of responsibility.